Introduction to MLOps

Vladislav Goncharenko

girafe ai

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Vladislav Goncharenko

- Author of machine learning courses and Masters program at MIPT
- Associate professor at MSU, HSE, Harbour.space
- ML researcher (MIPT)
- Team lead of video ranking team at Dzen (yandex.ru)
- Ex-team lead of perception team at self-driving trucks (Evocargo)
- Open source fan

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МФТИ

Дзен

Team

- Eugene Astafurov contributed a lot to materials and course structure
- Vladislav Naumov is seminarist and assistant

Preface

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Prerequisites

- UNIX command line basics
  - git basics
- Python programming experience
  - web services is a plus
- Basic machine learning knowledge
  - sklearn
  - pytorch

Disclaimer

- We discuss broad way to do things
- However there are plenty of ways
- We prefer ways suitable for most ML engineers
  - meaning not very deep software engineering skills
- Python is main tool for this course
  - MLOps exists in other stacks too, but for ML it’s mostly Python

Your part

1. Knowledge level of each student is different
2. We adjust materials based on your knowledge
3. Your knowledge is estimated by feedback

More you ask - more you get from this course!

I can give you a way but you need to make an effort to follow it

Outline

- Motivation
- What is MLOps?
- Subtasks
  - course structure
- Stacks overview
- Homeworks outline

Motivation for MLOps

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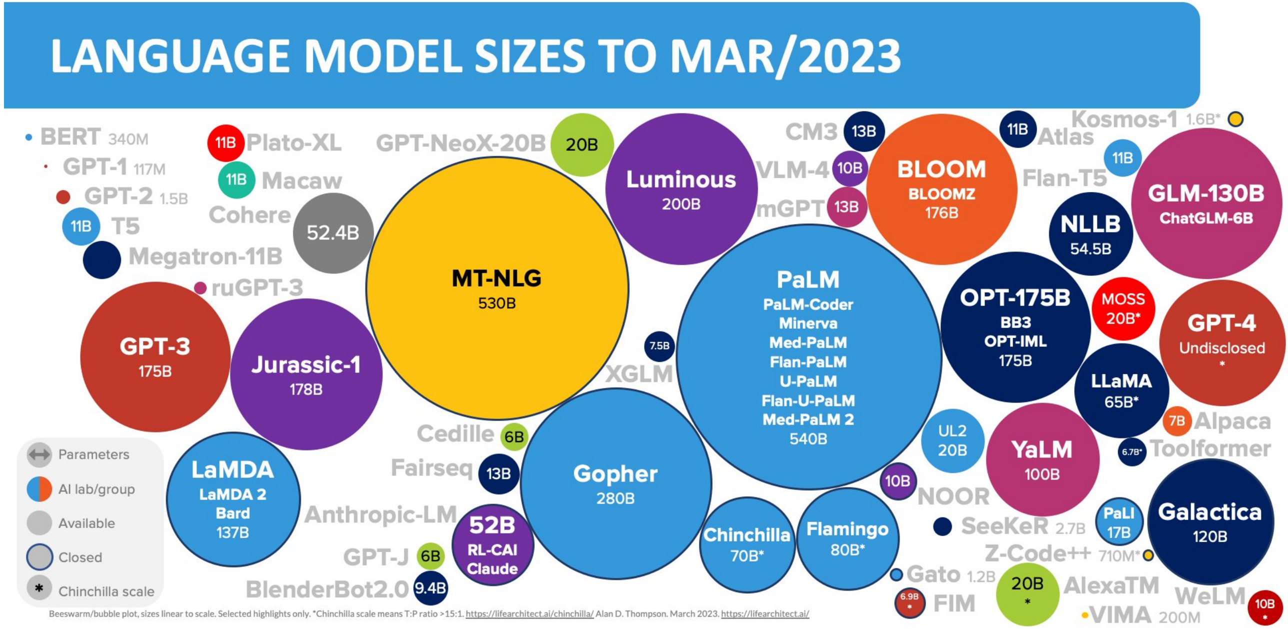
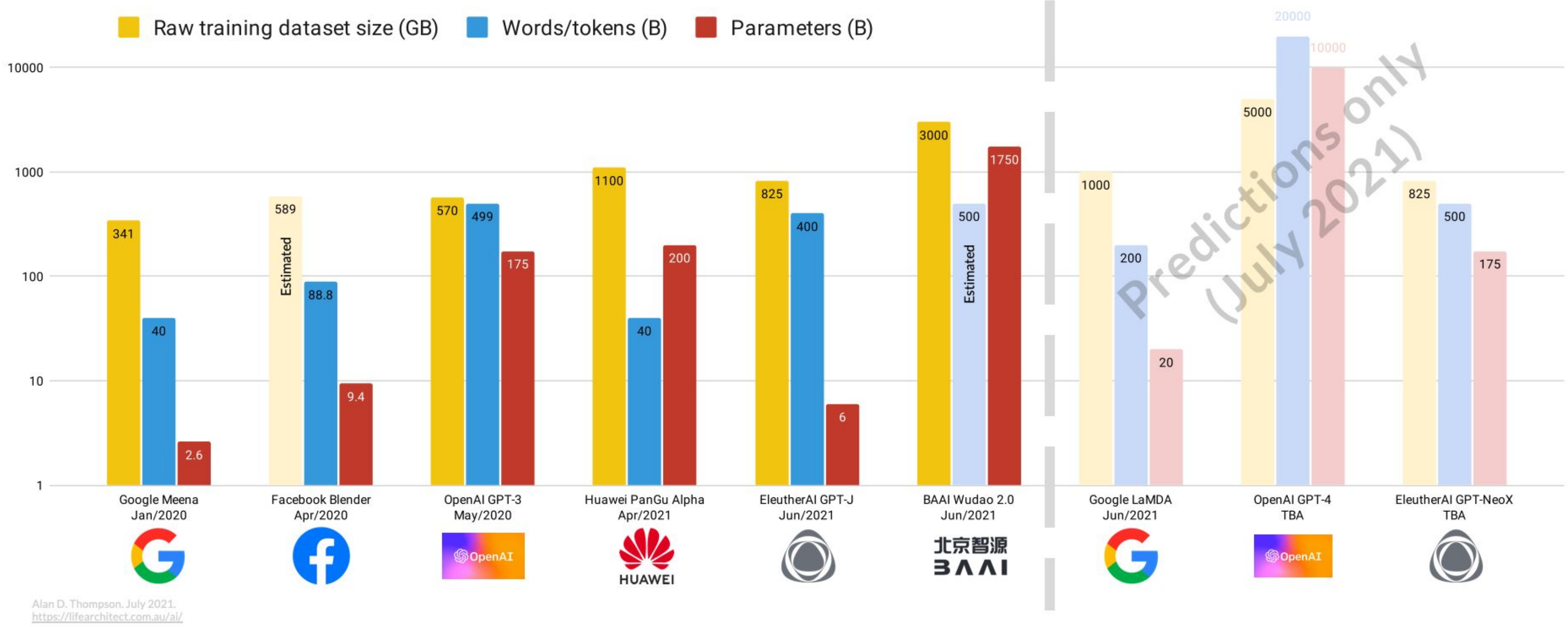
Why MLOps is needed?

- Resources for ML are growing
  - models training (gpu)
  - data mining (cpu, disks)
  - labeling (people)



LANGUAGE MODEL SIZES & PREDICTIONS (JUL/2021)



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  - models training (gpu)
- Results reproduction
  - not only for industry, but for academia as well

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- Results reproduction
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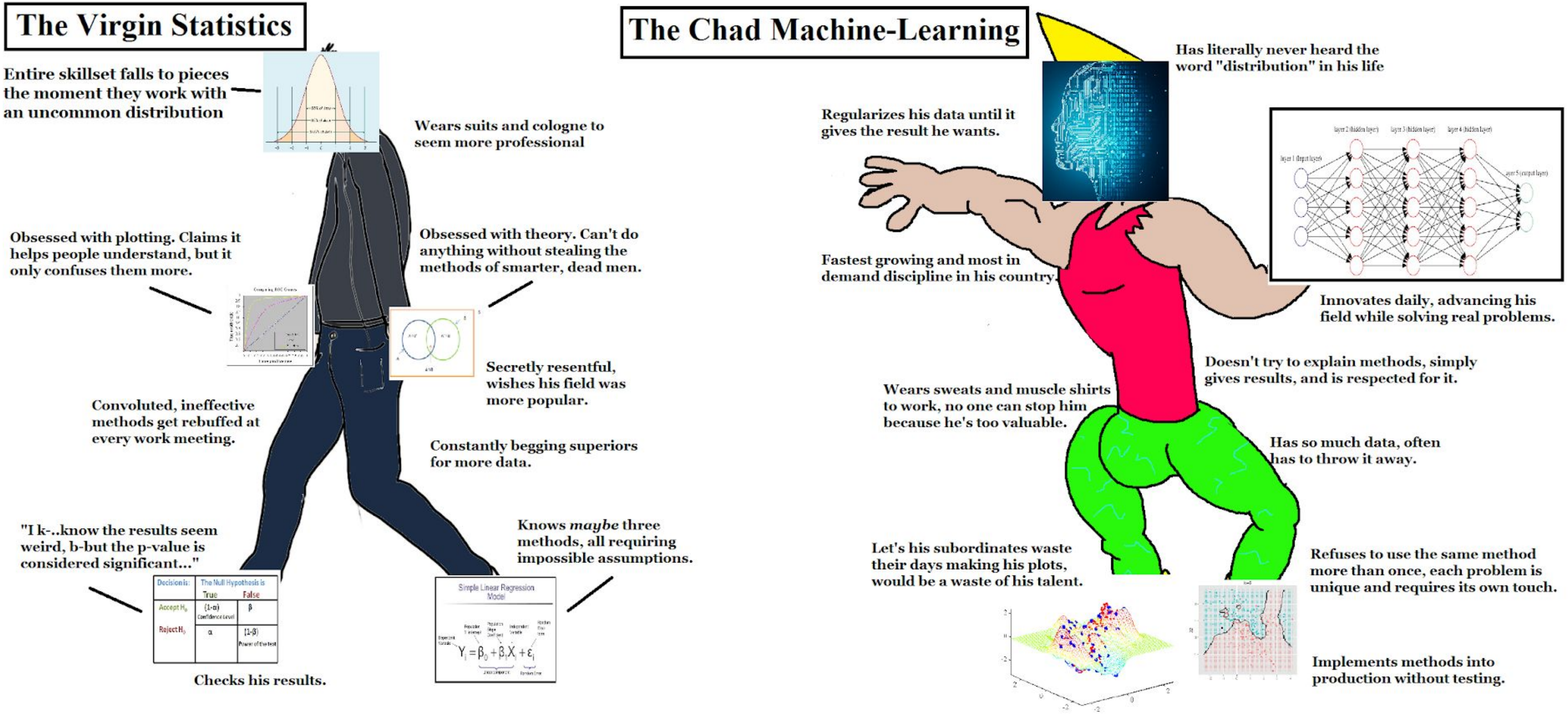
- Models delivery
    - reduce time to market
    - reduce routine
    - testing and reliability
- Competence decomposition
    - deeper division of labor

Why do ML specialists need MLOps?



Why do ML specialists need MLOps?

because no one will do it for you!!!

For deeper dive I suggest reviewing
system design primer - cookbook for modern web based systems

What is MLOps?

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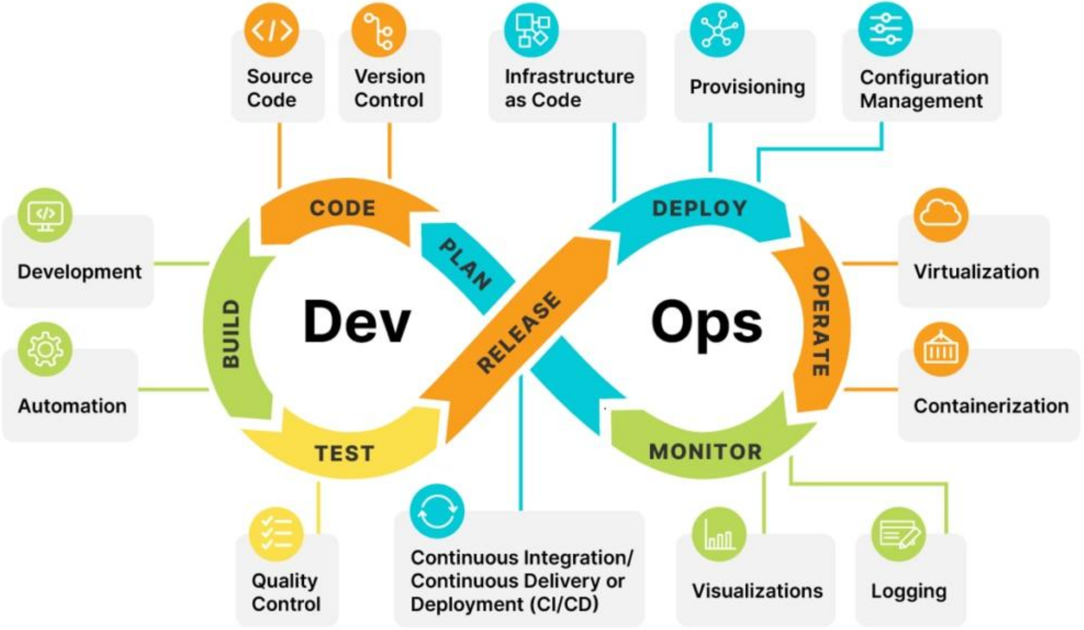
DevOps definition

DevOps is a methodology in the software development and IT industry. Used as a set of practices and tools.

DevOps integrates and automates the work of software development (Dev) and IT operations (Ops) as a means for improving and shortening the systems development life cycle

[Well-known common knowledge site]

[Great talk on DevOps practices]



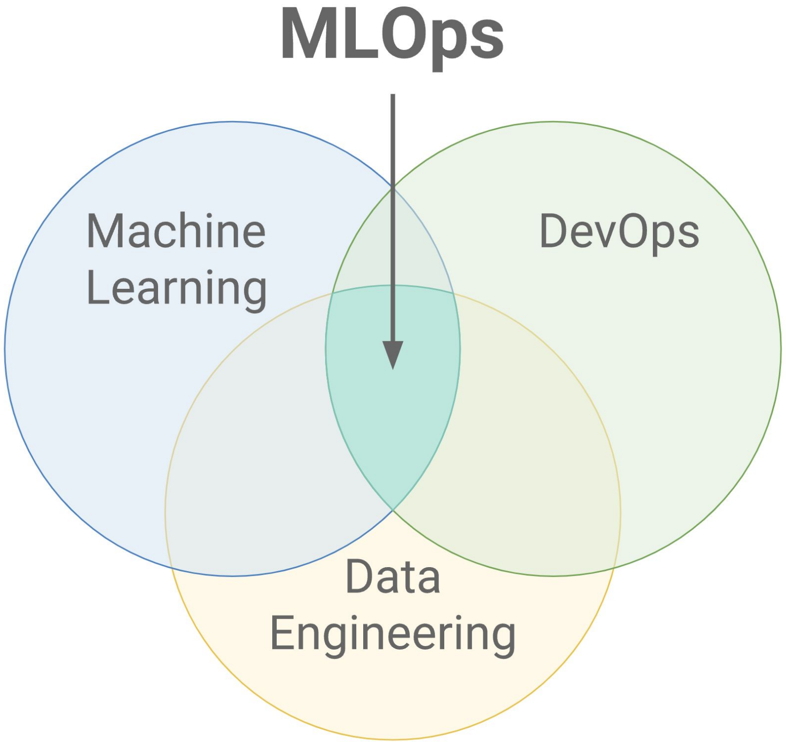
[DevOps course at MIPT (ru)]

MLOps definition

MLOps is a paradigm that aims to deploy and maintain machine learning models in production reliably and efficiently.

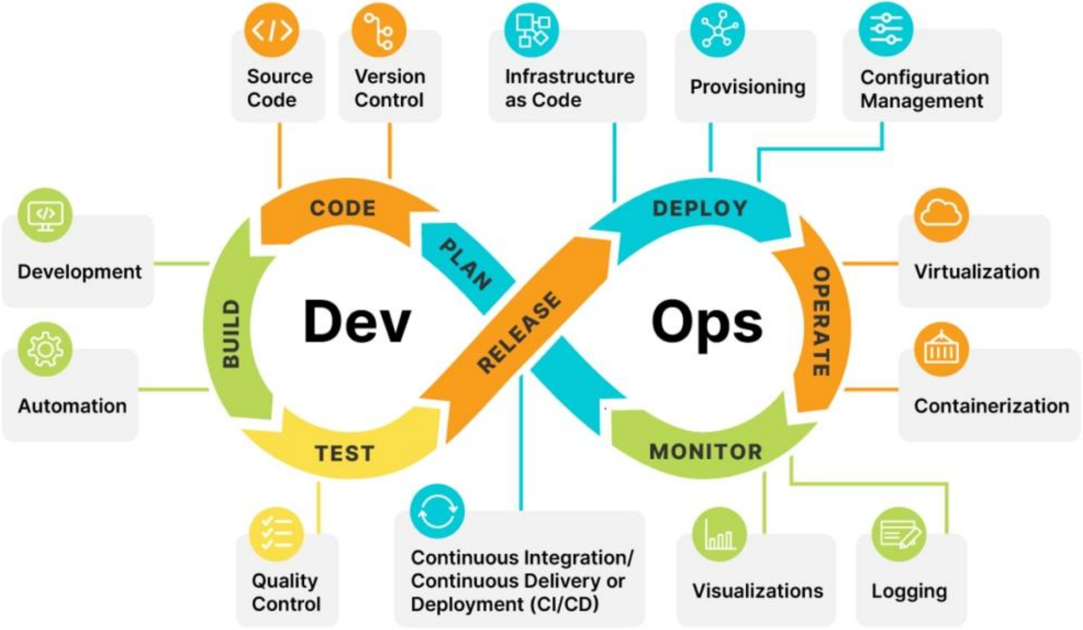
MLOps seeks to increase automation and improve the quality of production models, while also focusing on business and regulatory requirements

[Well-known common knowledge site]



Relations to other spheres

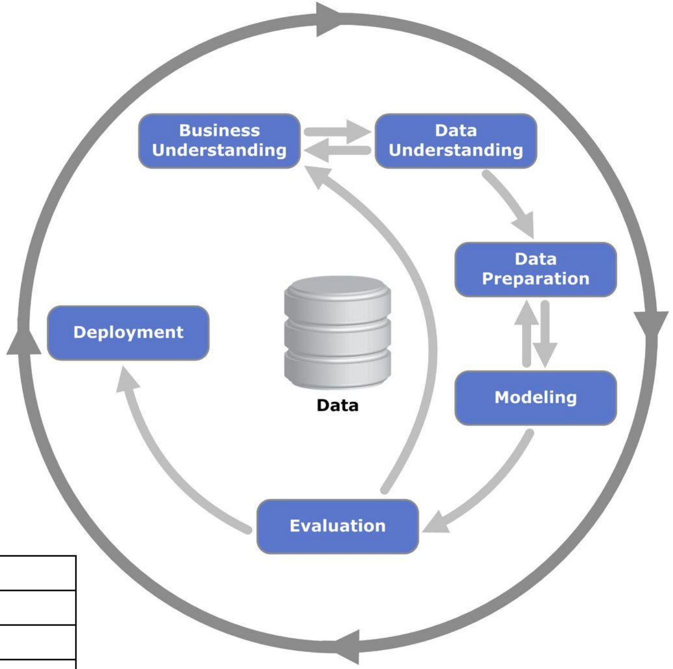
- Management
- System Design
- Software development
- SQL
- Analytics
- Data bases
- Machine Learning



CRISP DM

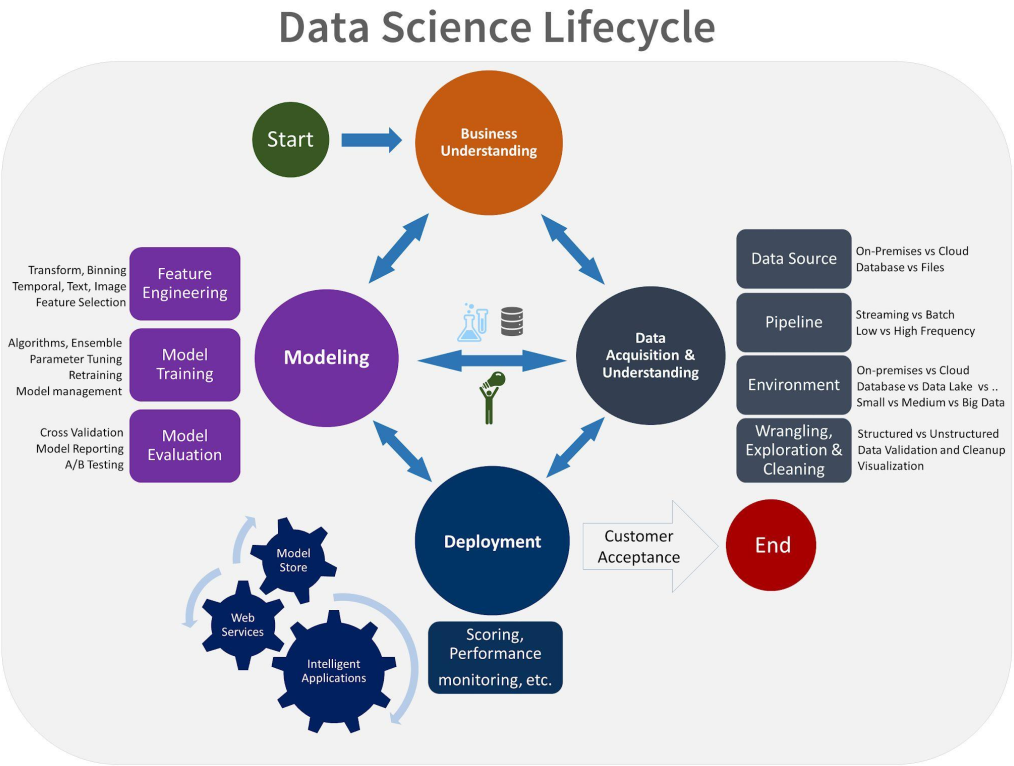
In the field of Data Science is used. **Cross-industry standard process for data mining** (CRISP DM) is a process standard for projects using predictive analytics, proposed in 1999. It was used by half of the companies engaged in Data Mining at the beginning of the century.

Read more



Poll Years	2002	2004	2007	2014
CRISP-DM	51%	42%	42%	43%
SEMMA	12%	10%	13%	8.5%
KDD process			7%	7.5%
My organization's	7%	6%	5%	3.5%
My own	23%	28%	19%	27.5%
Other (incl. domain specific)	4%	6%	9% (5%)	10% (2%)
None	4%	7%	5%	0%

Team Data Science Process

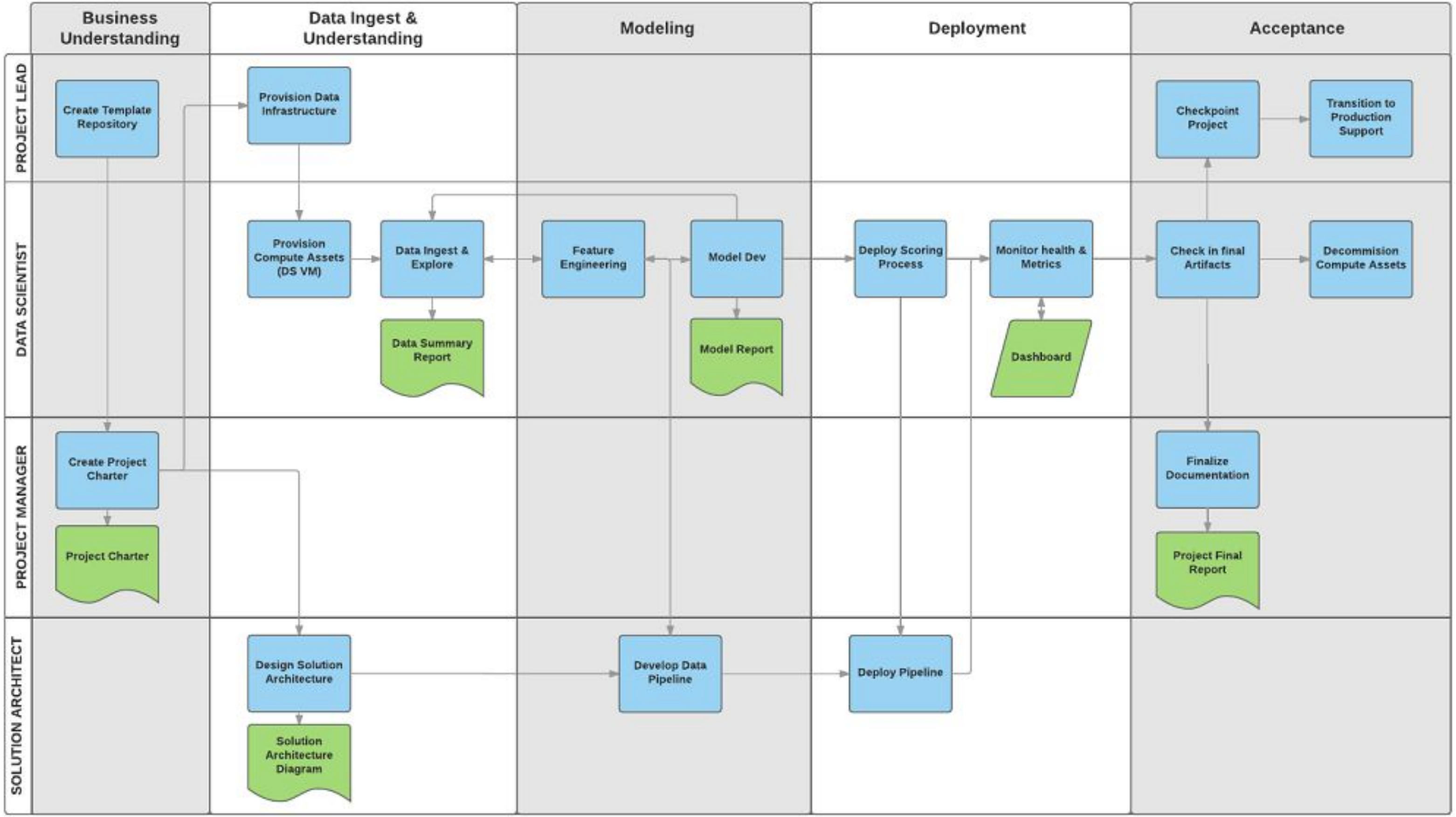


Modern standard from Microsoft

https://learn.microsoft.com/en-us/azure/architecture/data-science-process/overview

It is distinguished by the allocation of roles and a more detailed study of the stages of the cycle

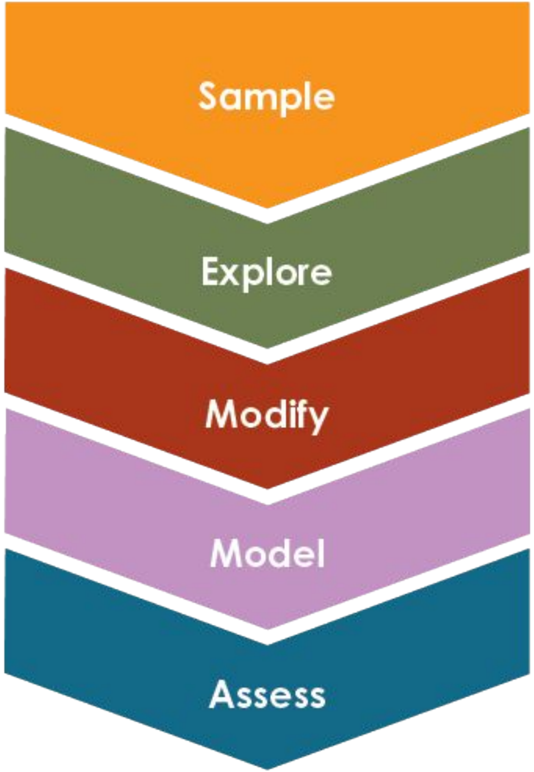
TDSP: team roles



Other approaches

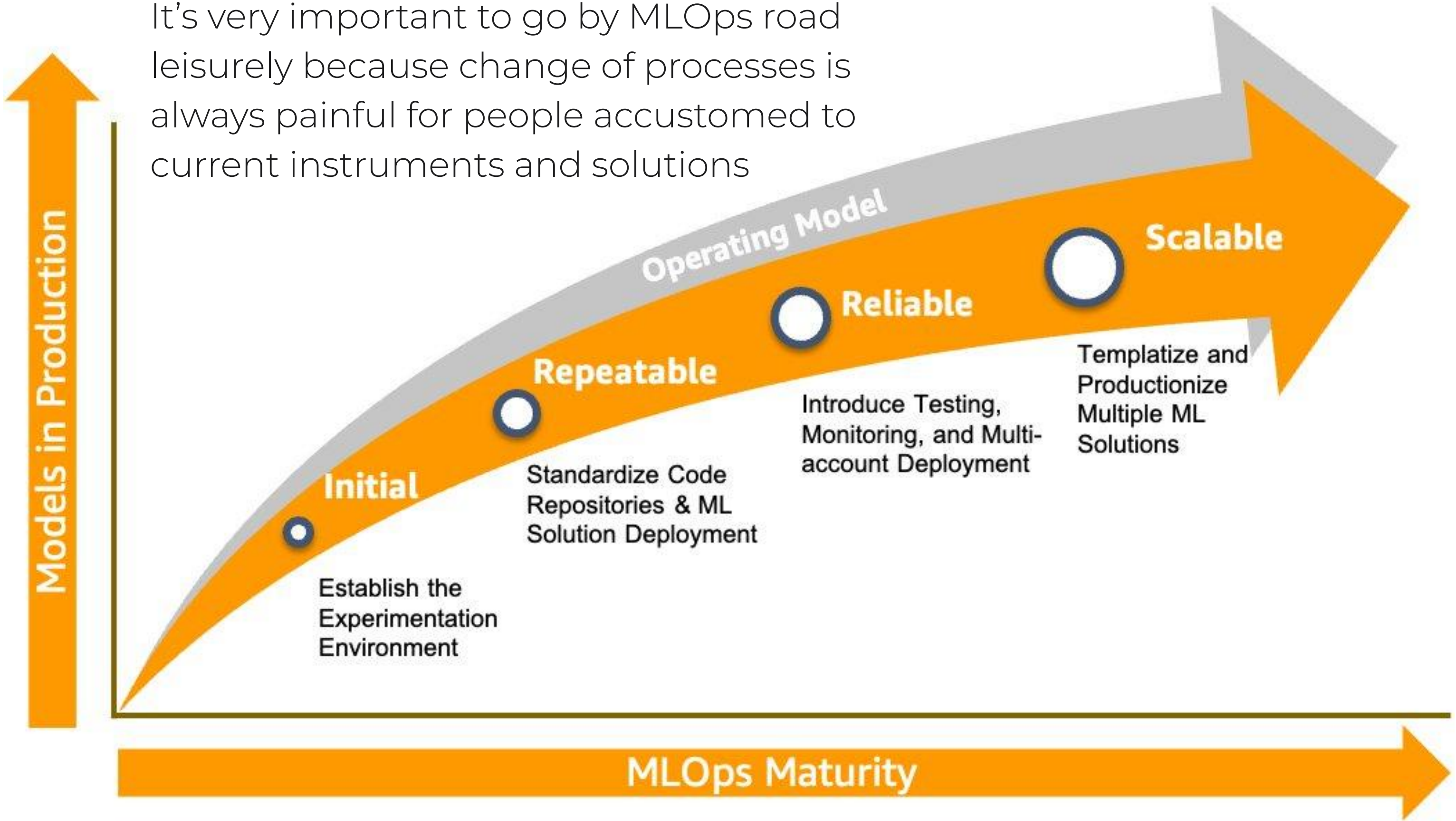
1. SEMMA
   read more
2. KDD Process
3. Waterfall
4. etc…

In fact these are topics for IT management course



MLOps gradual introduction steps

It’s very important to go by MLOps road leisurely because change of processes is always painful for people accustomed to current instruments and solutions



MLOps subtasks

Course structure

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MLOps subtasks

- Contemporary solutions knowledge
- Software engineering
  - Python infrastructure
- Virtualization
  - Docker
- Code storage and management
  - Git
  - CI & CD
  - code quality
  - testing (auto tests, test stands)
  - documentation generation
- Data storage and management
  - Data Acquisition
  - Data Engineering
  - AB tests

MLOps subtasks

- Contemporary solutions knowledge
- Software engineering
- Virtualization
- Code storage and management
- Data storage and management
- Labeling and crowdsourcing
- Experiments tracking
    - production monitoring
- Computational resources
    - GPU
- Model storages and deployment preparation
- Efficient inference
- Embedding storage optimization and vector databases

MLOps stacks overview

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MLOps at big tech

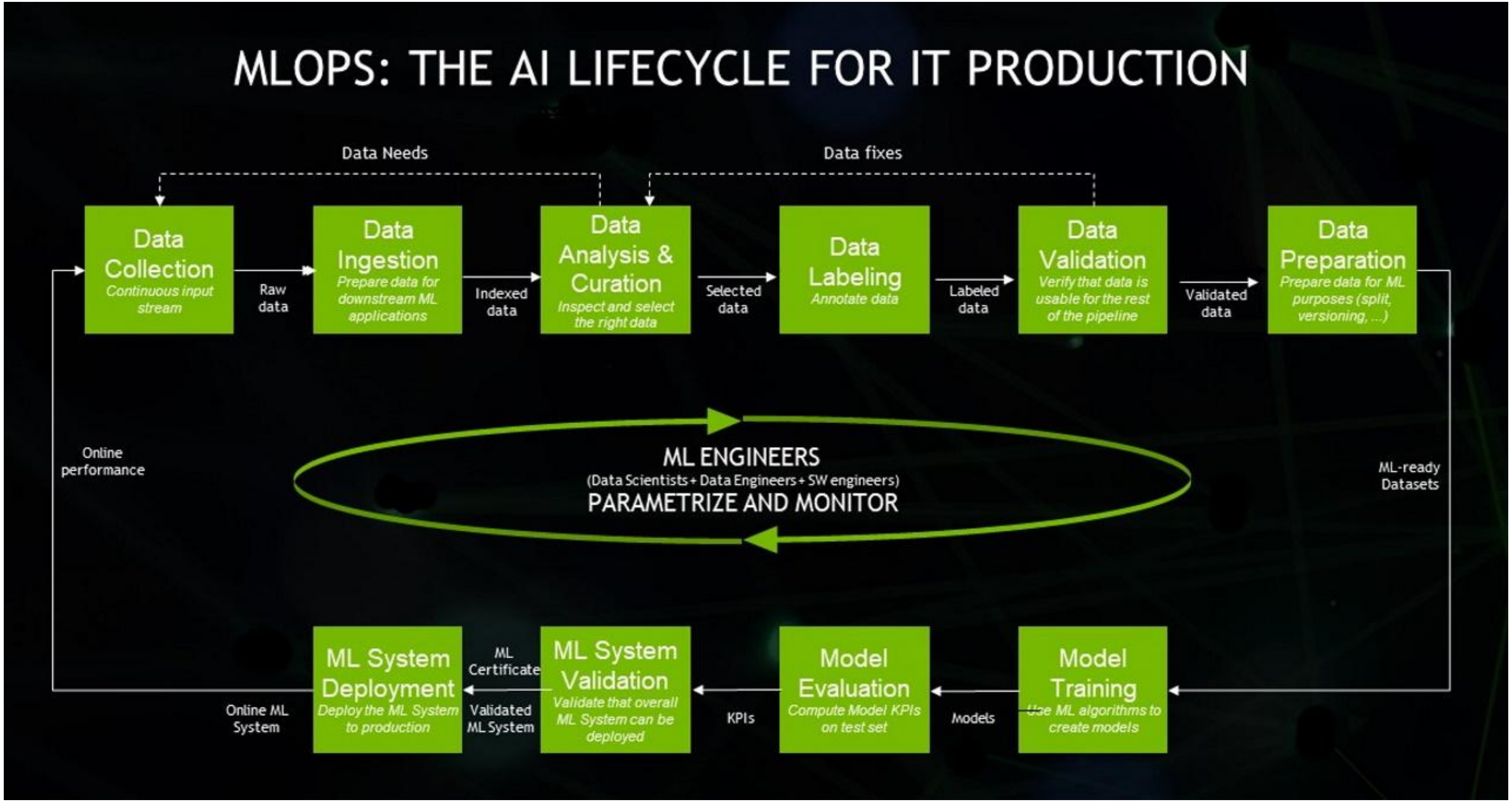
Big companies has their stack to organize models training and publishing pipelines

- Amazon SeageMaker
- Google Cloud
- Nvidia
- Microsoft Azure
- Casual clouds
- Yandex: YTSaurus + Nirvana + Toloka
- Databricks
- Seldon

Generally variance of instruments is huge over different companies (from only bare metal ssh machines to very solid custom solutions)

Nvidia

https://blogs.nvidia.com/blog/what-is-mlops/



Not only big tech

[Youtube video]

MLOps at academia

It just doesn’t exist (unfortunately)

You publish or perish:
no time for bullshit

What about open source stack?

Will discuss it during all the course.

Main problem of FOSS instruments - **lack of orchestration**.

Each instrument primarily aims to solve one task but then start to add some default solutions for adjacent problems which are not that great.

All in all no instrument solve every problem.
Finally you need to **write glue code yourself**.

Homeworks

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Homeworks

1. Formulate problem to apply MLOps stack
2. Write baseline code
3. Apply parts of pipeline from classes
4. Peer review
5. Final review

Also we will have simple tests for each lecture

Revise

- Motivation
- What is MLOps?
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  - and course structure
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Thanks for attention!

Questions?

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